

CHAPTER 6

ATTENTION MECHANISM BASED DEEP LEARNING MODELS FOR IMPROVING THE ACCURACY OF WATER QUALITY PREDICTION IN AQUACULTURE

This chapter investigates the development and implementation of attention mechanism-based DL models for enhancing the accuracy of WQP in aquaculture. To address this, ABi-LSTM and ABi-GRU DL-RNN models are proposed, specifically tailored for aquaculture WQP. The proposed models excel in their fusion of two essential elements: the attention mechanism's weighting feature and the non-linear mapping capability of Bi-LSTM and Bi-GRU. This unique combination allows the proposed DL-RNN models to offer enhanced accuracy and efficiency in WQP, making them a promising solution to the limitations faced by traditional methods. Section 6.1 of this chapter serves as an introductory section, outlining the key contributions of this work. Section 6.2 of this chapter explains the proposed ABD-RNN models. A critical aspect of building an accurate DL WQP model is the identification of the h_p . Section 6.3 presents an extensive study analysing the impact of various h_p variations on the performance of the proposed ABi-LSTM and ABi-GRU DL-RNN models. To facilitate a comprehensive comparison, two different WQ datasets are utilized for the experimental evaluations, considering existing models such as LSTM, GRU, Bi-LSTM, Bi-GRU, and A-LSTM and A-GRU. Section 6.4 delves deeper into the performance analysis of all the DL-RNN models for a selected set of h_p , ultimately presenting the final results and comparisons. The chapter is summarized in Section 6.5.

6.1 INTRODUCTION

The proliferation of DL technologies has led to more sophisticated mechanisms to parse and interpret complex patterns in data. A prime example is the attention mechanism, a feature central to this chapter, which focuses on its deployment within WQP. With proven efficacy in various DL tasks, the attention mechanism assigns differential attention or weights to distinct parts of the input data, thereby directing the model to focus on the most salient features.

This functionality proves instrumental when managing time series data, such as

those in WQP. It enables the model to determine the significance of different time steps in the prediction process, capturing essential temporal patterns within the data. Attention mechanism based RNN models, specifically attention-based LSTM and GRU models, are introduced to leverage these advantages. These models are further optimized by incorporating bidirectional LSTM/GRU layers with the attention mechanism, thus improving WQP. The efficacy of these models is critically assessed on the ADAK and MAC datasets, highlighting the significance of attention mechanisms in enhancing both model accuracy and computational efficiency.

Attention-based DL-RNN models have proven valuable in various time-series prediction tasks. For instance, in [124], a self-attention-based time-varying prediction model with a two-stage training method is proposed to improve performance compared to baseline DL-RNN models. Similarly, [125] introduces an integrated multiple directed attention (IMDA) DL-RNN model for air pollution prediction, demonstrating superior performance over other DL models. Furthermore, in [126], a self-attention-based forecasting model for regional influenza prediction (SAIFlu-Net) outperforms baseline DL-RNN models.

In order to meet the requirement of high prediction accuracy for high-intensity aquaculture, ABD-RNN models are proposed in this chapter. Bi-LSTM and Bi-GRU incorporate additional training layers that allow input data to be trained twice to improve the accuracy of predictions. In this work, the prediction accuracy and training time of the proposed ABi-LSTM and ABi-GRU models are compared with LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM and A-GRU DL-RNN models for WQP in aquaculture. The significant contributions of this work are as follows:

1. ABD-RNN (ABi-LSTM and ABi-GRU) models are proposed. The bi-directional models process the data in both forward and backward directions, enabling the model to capture more patterns than the unidirectional model along with the attention mechanism.
2. Attention mechanism is implemented with bidirectional models to take advantage of the weighing feature of the attention mechanism.
3. Extensive study of the impact of h_p on the performance of the proposed ABD-RNN models using the two different WQ datasets is presented. Determining the right set of hyper-parameters has a significant role in building an accurate DL-RNN WQP model.
4. The performance of ABD-RNN (ABi-LSTM and ABi-GRU) models are compared with baseline (LSTM and GRU), bidirectional (Bi-LSTM and Bi-GRU),

and attention-based (A-LSTM and A-GRU) DL-RNN models. Results show that proposed models significantly improve aquaculture WQP accuracy and are computationally efficient.

6.2 PROPOSED ABD-RNN MODELS

WQP is a typical time series problem, which can be considered to be a sequential machine learning regression problem. For training the model, this sequential time series data is converted into a set of feature values and the corresponding target value using the sliding window procedure since the regression is a supervised learning problem. The predicted output depends on the current input and the entire length of the time sequence with different levels of importance or weightage. Many prediction models map relationships with the input and output data at a particular time, ignoring the features in the lengthy sequence. To capture the features from this whole sequence data, bidirectional RNNs are used in the DL layer. This ensures the utilization of the information from the sequence data in the forward and backward directions. In the bidirectional NNs, with their non-linear mapping of the sequence, the DL layer accurately captures the sequence correlation of data to provide more precise prediction results. The DL layer is followed by an attention mechanism, which selectively focuses on and extracts the critical features from the sequence by giving the suitable weightage, thus improving the performance of the WQP model. The overview of the proposed ABD-RNN models is shown in Fig. 6.1.

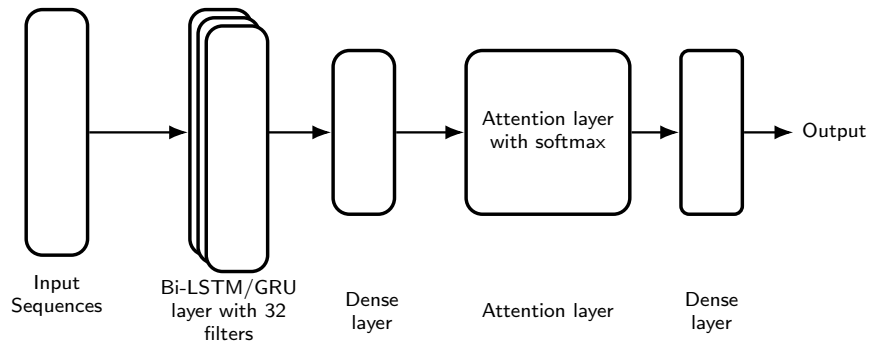


Figure 6.1: The proposed ABD-RNN model architecture

6.2.1 Bidirectional LSTM/GRU DL-RNN layer

In this DL layer, bi-directional DL-RNN (LSTM/GRU) is used. LSTM and GRU are two types of DL-RNN models used for sequence modeling tasks. LSTM was introduced to enhance long sequence learning and avoid the vanishing-gradient issue of RNN [138]. And GRU DL-RNN was proposed by authors [139] in 2014. It has per-

formance comparable with LSTM in terms of accuracy and lower demand for processing resources. Both models are designed to address the vanishing gradient problem in traditional RNNs, where the gradients of the error function can become infinitesimally small as they are propagated back through time.

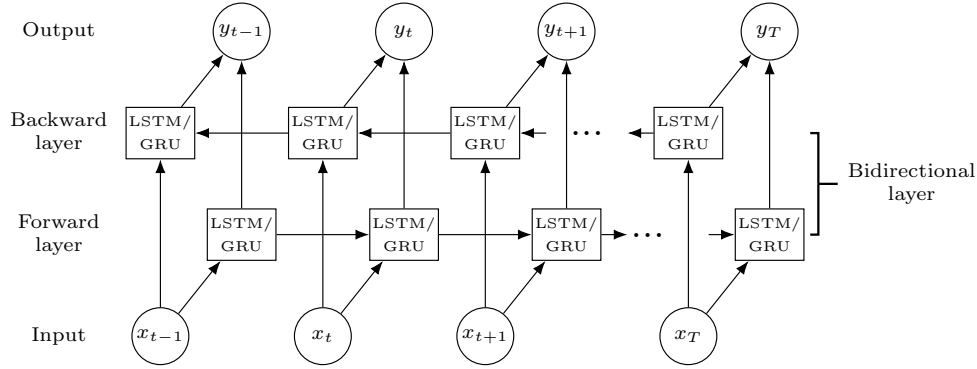


Figure 6.2: The Bi-LSTM/GRU structure

The bi-directional DL-RNN (LSTM/GRU) structure is shown in Fig. 6.2. The bidirectional DL models consist of two processes in both forward and backward directions, as depicted in Fig. 6.2. The processing in both directions helps the model capture patterns that the unidirectional LSTM and GRU models cannot capture. The bi-directional DL-RNNs are an extension of LSTM and GRU. Bi-directional DL-RNNs can fit data in both forward and backward directions and concatenate the prediction, while LSTM/GRU can only fit data in one direction [143]. A Bi-LSTM/Bi-GRU model first feeds data into one of the LSTM/GRU models for training the given data sequence, i.e., feedback layer and re-training with another LSTM/GRU model, but with the opposite order of the input data sequence. Bi-LSTM/Bi-GRU model can capture patterns ignored by LSTM/GRU because it has added a reverse directional layer. Double LSTM/GRU application improves long-term study and thereby improves accuracy [144]. Bi-directional networks are the extension of the LSTM/GRU model, which can learn about the relation of past and future factors of load and current load, which helps in extracting the in-depth features of loaded data [145].

6.2.2 Attention layer

After the Bi-LSTM/Bi-GRU layers extract the features in both directions, the output is fed to the attention layer [149]. The attention mechanism emphasizes the influence of critical segments by allocating distinct weights to hidden layer elements. The attention layer automatically understands the significance of each hidden state and assigns a weight to it based on its importance. The attention layer calculates the importance of

the input features by taking the dot product of the hidden state vector and a learned weight vector. The softmax function is then applied to ensure that the sum of all weights is 1, and the weights and the input features are multiplied and summed to get the output.

$$a_{tk} = \frac{\exp(e_{tk})}{\sum_{i=1}^n \exp(e_{ti})} \quad (6.1)$$

To implement the attention layer, first the attention weights a_k are calculated using the Eq. (6.1). Then multiply the attention weights by the hidden states and sum the result to get the output z_t . The learned variables v_e , b_e , W_e , and U_e are updated during training to improve the model's performance. First, for each time step t and each hidden state h_t , the attention layer calculates a score e_{tk} that represents the importance of h_t for predicting the value of the k -th input feature. This is done using the Eq. (6.2).

$$e_{tk} = v_{eT}(W_e[h_{t-1}, c_{t-1}] + U_e h_t + b_e) \quad (6.2)$$

In Eq. (6.2), v_{eT} , W_e , U_e , and b_e are learned parameters. v_{eT} is a row vector of size T , where T is the length of the input sequence. W_e is a matrix of size $T \times m$, where m is the number of neurons in the hidden layer. U_e is a matrix of size $m \times m$, and b_e is a bias vector of size $T \times m$. The vector $[h_{t-1}, c_{t-1}]$ is the concatenation of the previous hidden state h_{t-1} and cell state c_{t-1} , which are output by the LSTM or GRU layers.

The scores e_{tk} are then used to calculate the attention weights a_{tk} for each time step t and each input feature k . In Eq. (6.1), the numerator calculates the exponential of the score e_{tk} , while the denominator calculates the exponential of all scores e_{ti} for the same input feature k . The resulting attention weights a_{tk} represent the relative importance of each time step t for predicting the k -th input feature. The attention layer then computes the output z_t by taking a weighted sum of the hidden states h_t , using the attention weights a_{tk} .

$$z_t = \sum_{i=t}^T a_{ti} h_i \quad (6.3)$$

The Eq. (6.3) sums the attention weights a_{tk} for all time steps t from the current time step t to the end of the sequence T , and multiplies each weight by its corresponding hidden state h_t . The resulting output z_t is a weighted combination of all the hidden states, with higher weights given to the more important time steps as determined by the attention mechanism. Overall, the attention layer allows the model to selectively focus on the most relevant features and time steps for predicting the target variable, leading to improved prediction accuracy.

6.2.3 Model implementation

After standardization, a dataset is scaled to unit variance, removing the mean from the dataset. The whole dataset is normalized using the train set before training the DL-RNN models. The StandardScaler module from the Scikit-Learn Python library is utilized [146] to implement it in these DL-RNN models. The WQ sequences are broken into smaller segments once the datasets have been standardized by moving a window of the same size depending on the specified window size (W_z). By creating many short sequences from a single lengthy series, this step reduces the input sequence to a fixed-length sample and increases the sample size for training the model. This transformed data is used to train all the DL-RNN models.

6.3 EXPERIMENTS: HYPERPARAMETER TUNING

In this section, two WQ data sets are used to conduct experiments to analyse and evaluate the performance of the proposed attention-based bi-directional DL-RNN models (ABi-LSTM and ABi-GRU) models. Comparing the DL-RNN models with our proposed models in terms of prediction accuracy and computation time performance, the effectiveness of the proposed models can be validated. All experiments are conducted on Microsoft Azure Virtual Machines with specifications: Inter(R) Xeon (R) 8272CL CPU @2.60GHz, 32 GB RAM. The open source software used are Python and Keras based on Tensorflow.

The ABi-LSTM and ABi-GRU DL-RNN models are compared with LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, and A-GRU DL-RNN models. In the data set, 80% of the data is used to train the model, and the remaining 20% to test the model's prediction accuracy. The most crucial task in making the most effective and efficient DL-RNN model is setting h_p and optimizing them. In this work, the focus is to improve prediction accuracy without compromising the computation efficiency of the model. To find the model with the best prediction accuracy without compromising on computation efficiency, the performance of all the DL-RNN models are extensively studied through five sets of experiments varying each h_p . The fixed h_p of each model are dropout of 0.2 is applied after the primary layers to avoid over-fitting, and *ReLU* is applied as the activation function. The loss function used in all the models is MSE. The number of neurons of each hidden layer is set to 32.

The extensive study included the experiments with the tuning of following five h_p : optimizer (O_p), learning rate (L_r), epochs (E_p), window size (W_z) and batch size (B_z). The h_p are plotted against RMSE. The h_p are plotted against the computation time as well. All the experiments are done on both datasets.

6.3.1 Optimizers (O_p)

O_p are algorithms that update each layer's attributes, such as weights and L_r , to reduce the loss after each iteration. A suitable O_p determines the right weight and gives the best results [150]. The performance of DL-RNN models with various O_p is studied to find the ideal one for this work. The O_p used in this work are Adam, AdaGrad, AdaMAX, Nadam, Ftrl, and RMSprop. In Fig. 6.3 the RMSE bar plots for different O_p indicating the prediction accuracy of each DL-RNN model for various WQ parameters on the ADAK WQ dataset are shown. Compared to the proposed ABD-RNN models, it can be observed that the performance of the other DL-RNN models is inadequate.

Figure 6.4 shows the RMSE bar plots for different O_p to indicate the prediction accuracy of each DL-RNN model for various WQ parameters on the MAC WQ dataset. The bar plots show the ABD-RNN models have outperformed all other models. Adam produces the best results for each DL-RNN model. Hence Adam is chosen as the default O_p for all further experiments and the final comparison of the prediction performance of all DL-RNN models.

6.3.2 Learning rate (L_r)

This experiment investigate the prediction accuracy of each DL-RNN model for a defined set of L_r $\{0.01, 0.001, 0.0008, 0.0006, 0.0004, 0.0001\}$. Figure 6.5 shows the bar plot RMSE for different DL-RNN models for the defined set of L_r while tested with the ADAK WQ dataset. The proposed ABD-RNN models perform better than other DL-RNN models.

The same experiment is repeated with the MAC WQ dataset to review the performance of these DL-RNN models for a different dataset. Figure 6.6 shows the bar plots of RMSE for each DL-RNN model for various L_r with the MAC WQ dataset. Here, from Fig. 6.6 it can be observed that the bidirectional and attention-based models perform better when compared to baseline models. This experiment helps in selecting the default L_r for other experiments. From Fig. 6.6, it can be observed that all the DL-RNN models achieved their best prediction accuracy for each WQ parameter in both datasets when $L_r = 0.001$.

6.3.3 Epoch (E_p)

In this experiment, the outcome of choosing different E_p $\{10, 20, 30, 40, 50\}$ for each DL-RNN model is studied. Figure 6.7 shows the RMSE bar plots for various E_p for predicting each WQ parameter in the ADAK WQ dataset using different DL-RNN models. The proposed bidirectional DL-RNN models Bi-LSTM and Bi-GRU maintain

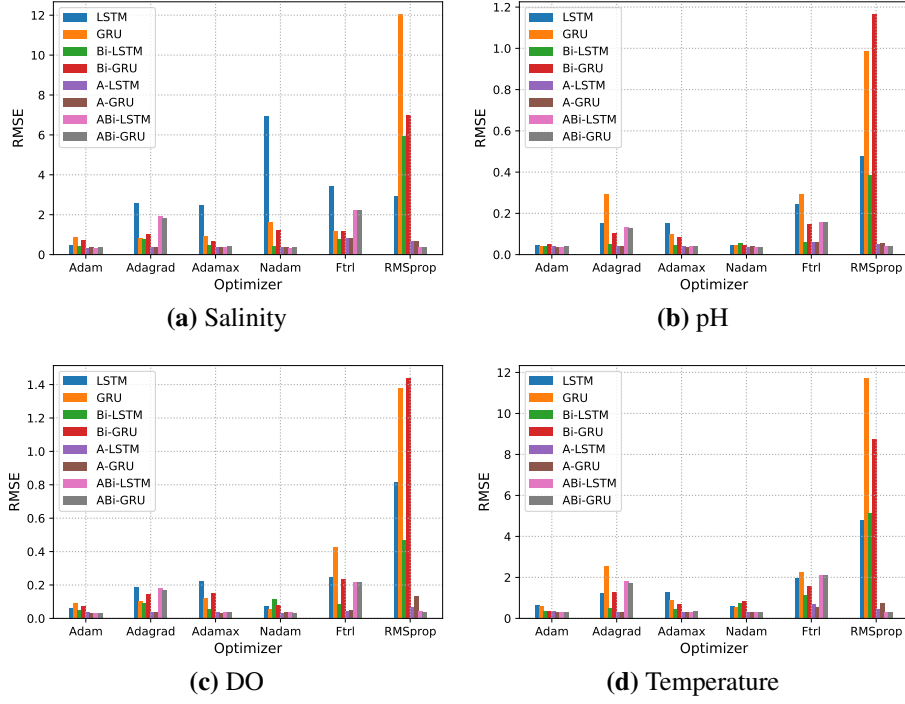


Figure 6.3: Comparison of prediction accuracy (RMSE) of DL-RNN models for various O_p and fixed h_p ($L_r = 0.001$, $E_p = 10$, $W_z = 30$, and $B_z = 64$) on the ADAK water quality dataset

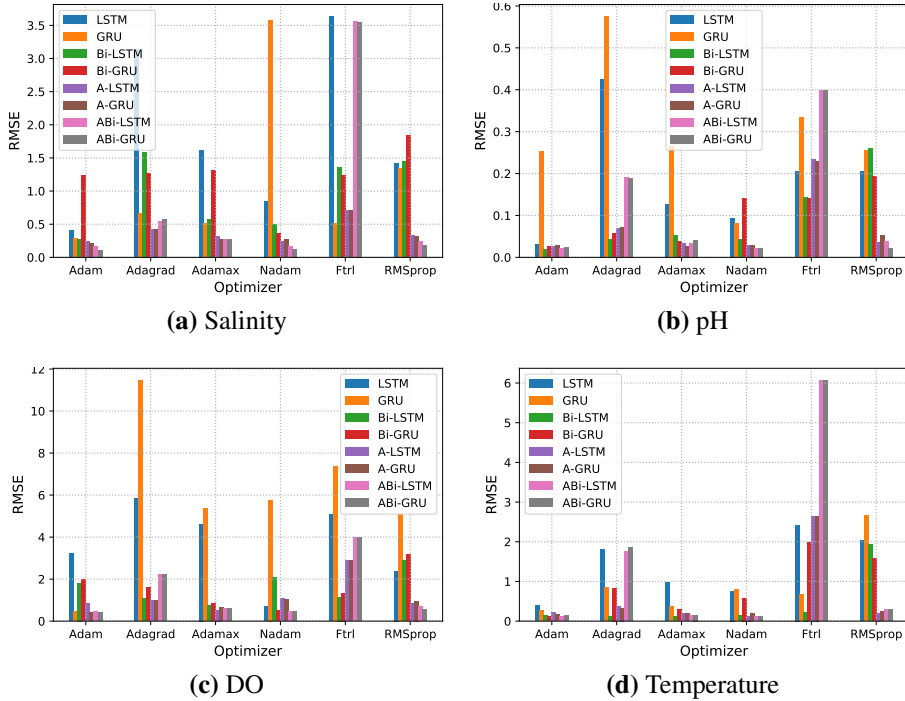


Figure 6.4: Comparison of prediction accuracy (RMSE) of DL-RNN models for various O_p and fixed h_p ($L_r = 0.001$, $E_p = 10$, $W_z = 30$, and $B_z = 64$) on the MAC water quality dataset

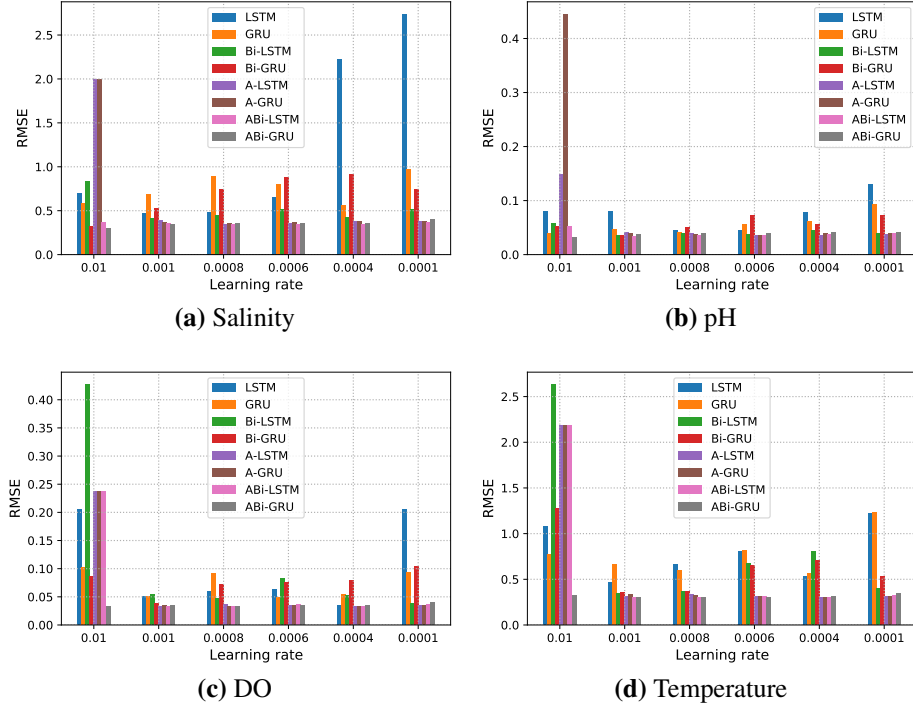


Figure 6.5: Comparison of prediction accuracy (RMSE) of DL-RNN models for various L_r and fixed h_p ($E_p = 10$, $W_z = 30$, $B_z = 64$ and O_p is Adam) on the ADAK water quality dataset

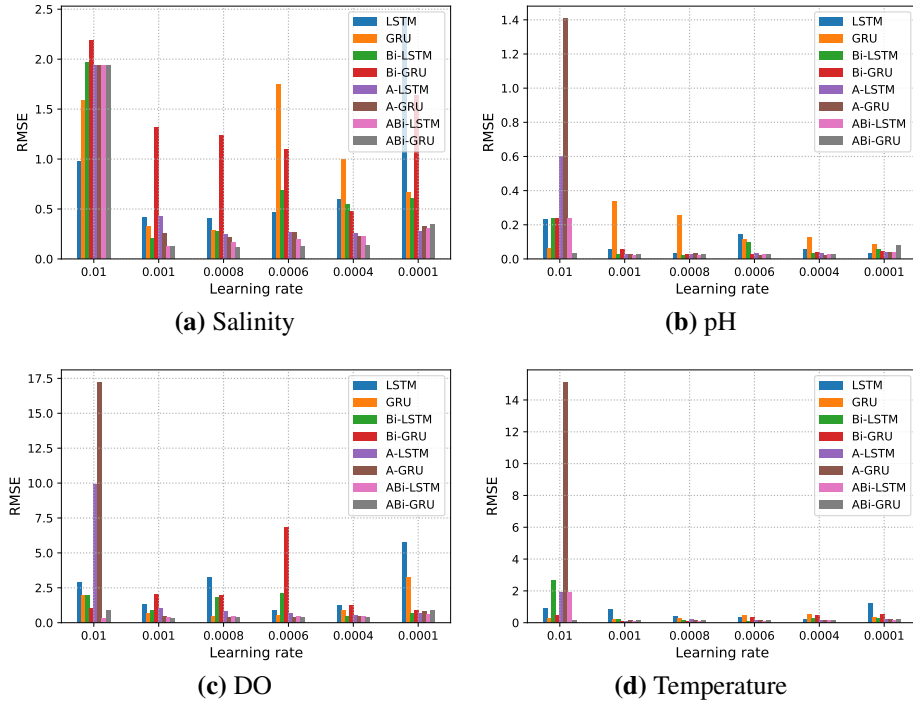


Figure 6.6: Comparison of prediction accuracy (RMSE) of DL-RNN models for various L_r and fixed h_p ($E_p = 10$, $W_z = 30$, $B_z = 64$ and O_p is Adam) on the MAC water quality dataset

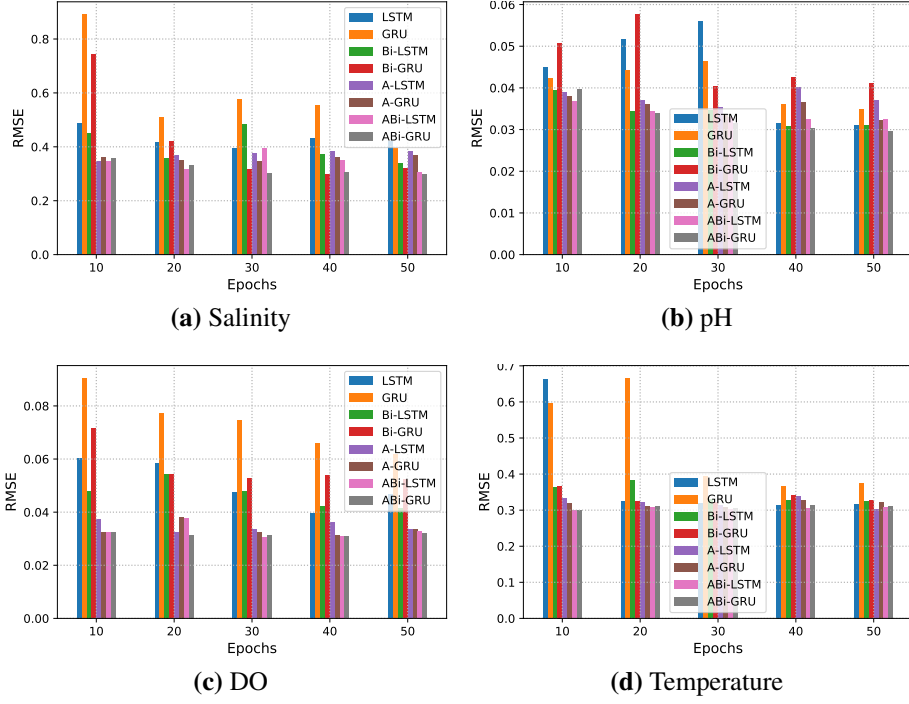


Figure 6.7: Comparison of prediction accuracy (RMSE) of DL-RNN models for various E_p and fixed h_p ($L_r = 0.001$, $W_z = 30$, $B_z = 64$ and O_p is Adam) on the ADAK water quality dataset

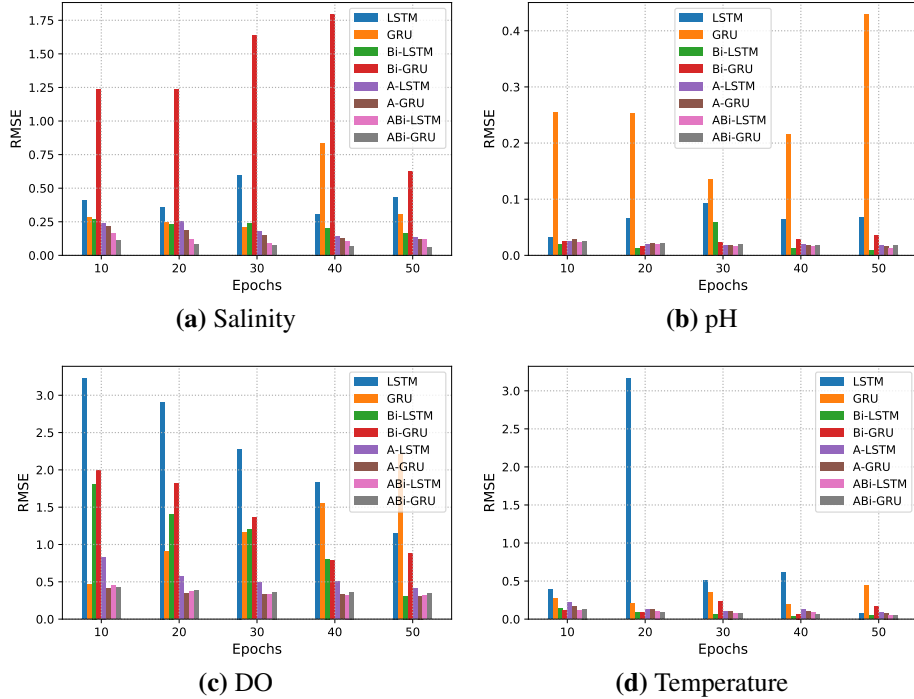


Figure 6.8: Comparison of prediction accuracy (RMSE) of DL-RNN models for various E_p and fixed h_p ($L_r = 0.001$, $W_z = 30$, $B_z = 64$ and O_p is Adam) on the MAC water quality dataset

better performance than baseline DL-RNN models LSTM and GRU. Bidirectional DL-RNN and attention-based DL-RNN models achieve good performance within $E_p = 10$ to 20, and the generalization is attained within $E_p = 20$. Prediction accuracy is maintained for the prediction performance of all four WQ parameters for $E_p = 10$ to 50. Also, the bidirectional and attention-based DL-RNN models are higher than the baseline DL-RNN models. Baseline DL-RNN models require more E_p to achieve generalization and are under-fitting training data for fewer E_p . The baseline models need more computation resources to achieve the prediction accuracy of the proposed ABD-RNN (ABi-LSTM and ABi-GRU) models.

The same experiment is done with the MAC WQ dataset to evaluate the performance further. The RMSE bar plots for various E_p assess the prediction accuracy of four WQ parameters of the MAC dataset with all DL-RNN models are shown in Fig. 6.8. Here the proposed ABD-RNN models perform much better than the baseline DL-RNN models for predictions of WQ parameters. The ABD-RNN models attained good accuracy and generalization within $E_p = 20$ in both datasets. Furthermore, the accuracy loss is minimal after increasing E_p . As the experiment is advanced, the $E_p = 10$ is chosen for all models since all the proposed models achieved the best prediction accuracy for all WQ parameters in both datasets. In the results section, the results on both datasets is compared with $E_p = 10$ and 50.

6.3.4 Window size (W_z)

W_z is also known as a look back. W_z is the number of samples used to predict one instance. In this experiment, the performance of each DL-RNN model while selecting different W_z is analysed. Figure 6.9 shows the RMSE bar plots for different W_z for predicting each WQ parameter on the ADAK WQ dataset using all DL-RNN models. The ABD-RNN models outperform all other models. The accuracy of the baseline models improves the W_z is increased, indicating baseline models require more resources for better prediction accuracy. In this experiment, baseline model accuracy is less than other DL-RNN models for different W_z .

The same experiment is done with the MAC WQ dataset to evaluate the performance further. The RMSE bar plots for various W_z assess the prediction accuracy of four WQ parameters of the MAC dataset with all DL-RNN models are shown in Fig. 6.8. Furthermore, it can be observed from Fig. 6.10 that the bidirectional models consistently perform well compared to the other models. As this experiment advances, the W_z can be chosen as 30 for all models as all the models achieved good prediction accuracy on both datasets. In addition, it can be observed from the experiments that the choice of W_z influences the prediction accuracy and computation time of all DL-RNN

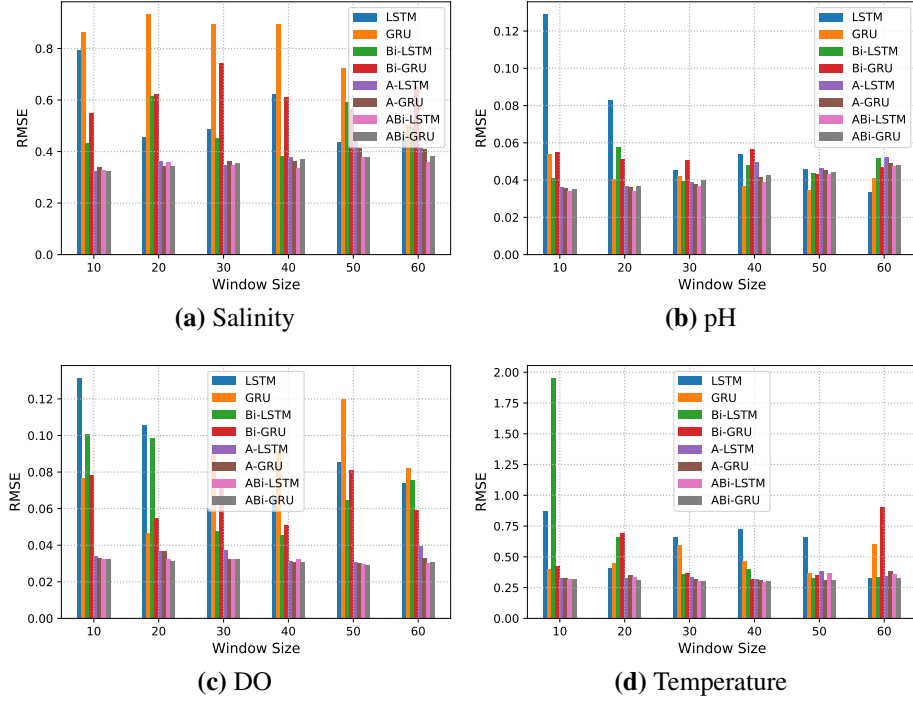


Figure 6.9: Comparison of prediction accuracy (RMSE) of DL-RNN models for various W_z and fixed h_p ($L_r = 0.001$, $E_p = 10$, $B_z = 64$ and O_p is Adam) on the ADAK water quality dataset

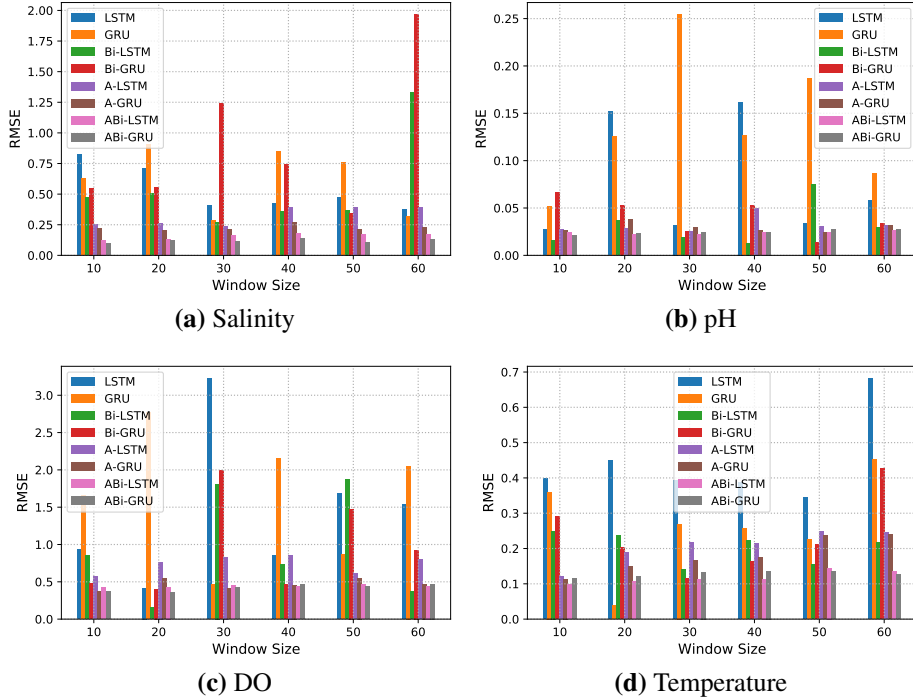


Figure 6.10: Comparison of prediction accuracy (RMSE) of DL-RNN models for various W_z and fixed h_p ($L_r = 0.001$, $E_p = 10$, $B_z = 64$ and O_p is Adam) on the MAC water quality dataset

models. As the W_z increases required computation time also increases.

6.3.5 Batch size (B_z)

B_z is the number of sub-samples that the NN uses to update the weights in each iteration and the right weight and gives the best results. The prediction performance of DL-RNN models with $B_z \{16, 32, 64, 128, 256, 512\}$ is analysed. In Fig. 6.11, the RMSE bar plots indicating the prediction accuracy of each DL-RNN model with different B_z for four WQ parameters in the ADAK WQ dataset are shown. Compared to the proposed ABD-RNN models, it can be observed that the performance of the other DL-RNN models is inadequate.

Figure 6.12 shows the RMSE bar plots indicating the prediction accuracy of each DL-RNN model with different B_z for four WQ parameters in the MAC WQ dataset. In this experiment, it can be observed that compared to ABD-RNN models, other DL-RNN models have less prediction accuracy for different B_z .

6.3.6 Computation time

While studying the prediction performance by varying O_p , L_r , E_p , W_z , and B_z , the computation time required for each model with each one of the h_p was also noted. Figure 6.13 and Fig. 6.14 show the bar plot for the computation time of each DL-RNN model for each set of h_p on the ADAK and MAC WQ dataset, respectively. From Fig. 6.13 (a) shows the computation time in seconds of each model for various L_r , it can be observed that the computation time is increasing with a decrease in L_r . From Fig. 6.13 (b), it can be observed that the computation time is increasing with an increase in E_p . A similar observation can be seen in Fig. 6.13 (c), where the computation time increases with an increase in W_z . In Fig. 6.13 (d), the computation time decreases with an increase in B_z . When considering O_p , it is evident that the computation time is the least for Adam O_p . This observation makes the selection of Adam a straightforward choice for all the experiments and in the results section. The bar plots in Fig. 6.14 of MAC data for computation time of various h_p help us conclude that the observation made in the ADAK data is valid.

6.4 RESULTS AND DISCUSSIONS

The objective is to finally analyse and report the prediction performance of the proposed ABD-RNN models and compare them with LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, and A-GRU DL-RNN models on both datasets. The performance comparison is made with a fixed set of h_p on the two WQ datasets. Each of the h_p is selected by studying the prediction performance and computation efficiency of the models for

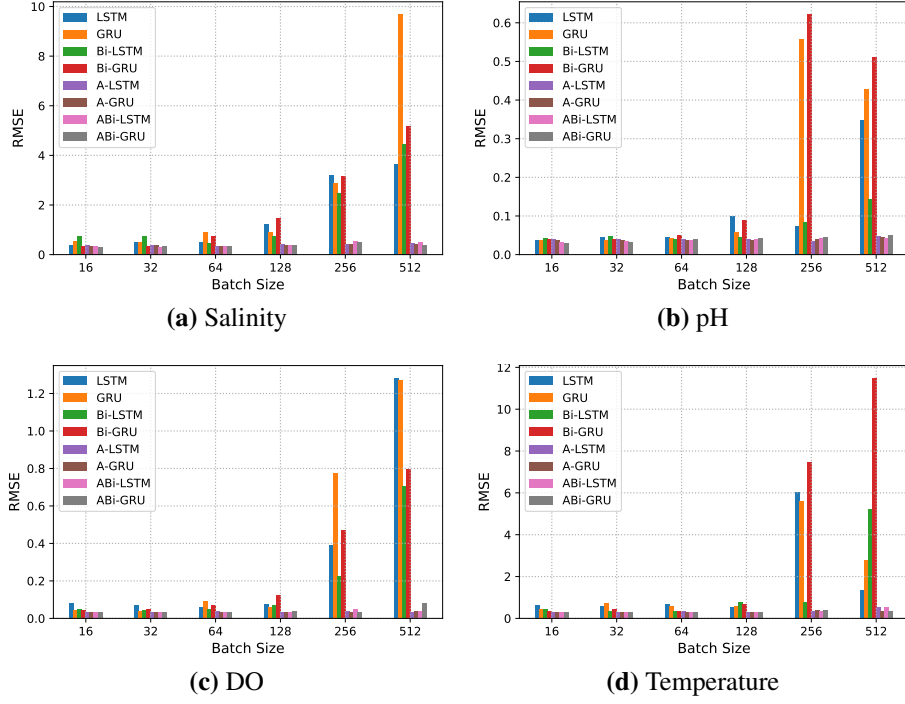


Figure 6.11: Comparison of prediction accuracy (RMSE) of DL-RNN models for various B_z and fixed h_p ($L_r = 0.001$, $E_p = 10$, $W_z = 30$, and O_p is Adam) on the ADAK water quality dataset

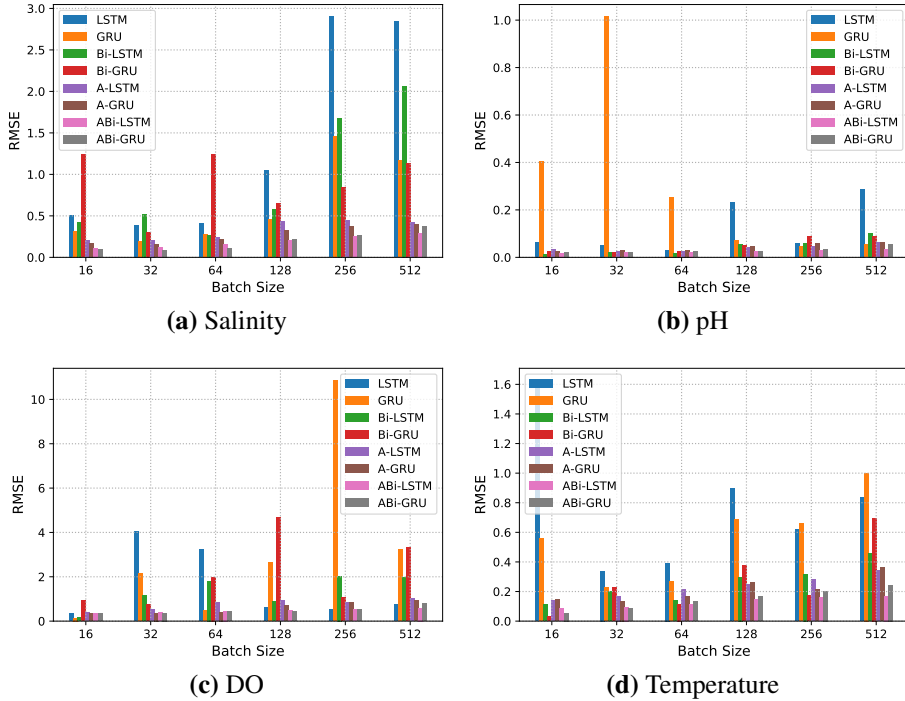


Figure 6.12: Comparison of prediction accuracy (RMSE) of DL-RNN models for various L_r and fixed h_p ($E_p = 10$, $W_z = 30$, $B_z = 64$ and O_p is Adam) on the ADAK water quality dataset

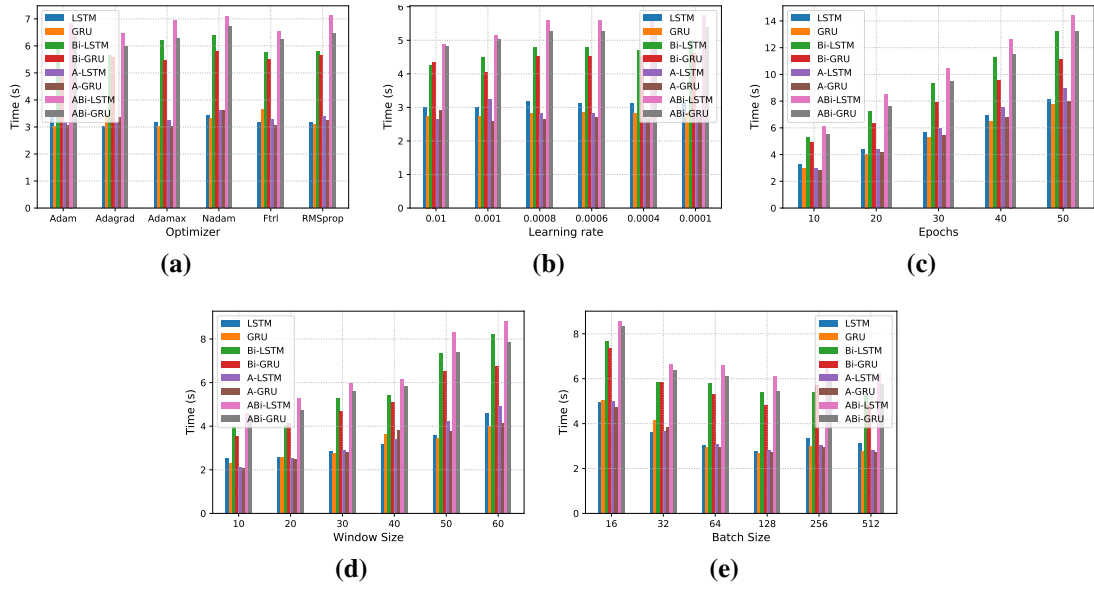


Figure 6.13: Comparison of computation time for prediction using all DL-RNN models. (a) $O_p = \{\text{Adam, AdaGrad, AdaMAX, Nadam, Ftrl, RMSprop}\}$, (b) $L_r = \{0.01, 0.001, 0.0008, 0.0006, 0.0004, 0.0001\}$, (c) $E_p = \{10, 20, 30, 40, 50\}$, (d) $W_z = \{10, 20, 30, 40, 50, 60\}$, and (e) $B_z = \{16, 32, 64, 128, 256, 512\}$ on ADAM water quality dataset

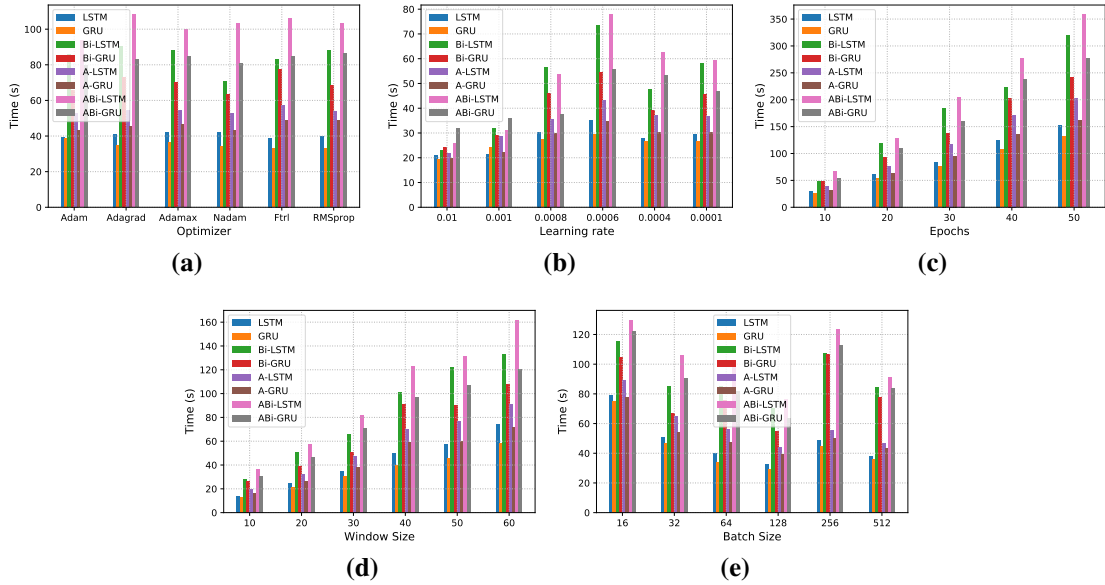


Figure 6.14: Comparison of computation time for prediction using all DL-RNN models. (a) $O_p = \{\text{Adam, AdaGrad, AdaMAX, Nadam, Ftrl, RMSprop}\}$, (b) $L_r = \{0.01, 0.001, 0.0008, 0.0006, 0.0004, 0.0001\}$, (c) $E_p = \{10, 20, 30, 40, 50\}$, (d) $W_z = \{10, 20, 30, 40, 50, 60\}$, and (e) $B_z = \{16, 32, 64, 128, 256, 512\}$ on MAC water quality dataset

each h_p (O_p , L_r , E_p , W_z and B_z) in the experiment section. This selected optimal set of h_p is used for comparing the performance of all the DL-RNN models in this section.

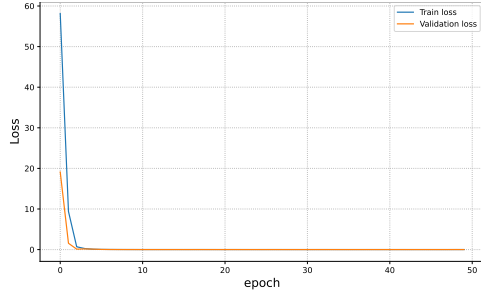
Figure 6.15 and Fig. 6.16 show the loss plots for convergence analysis of DL-RNN models. The h_p utilized in the training process include are $E_p = 50$, $L_r = 0.001$, $W_z = 30$, $B_z = 64$, and O_p is Adam on both the ADAK and MAC water quality datasets. Examining the learning curves for each model, it is evident that the training loss progressively decreases and stabilizes. Additionally, the validation loss exhibits a similar trend of decreasing and stabilizing, with a slight gap with the training loss. The combined loss plots suggest that the DL-RNN models are in a balance between being overfitted and underfitted, signifying their effective ability to generalize across the data.

6.4.1 ADAK water quality dataset

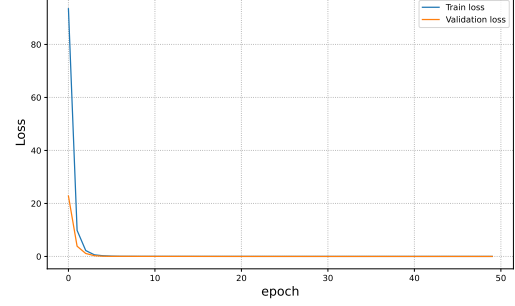
The prediction result of the ADAK WQ dataset is reported in Table 6.1, which gives MAE, MSE, RMSE and MAPE comparative analysis of the baseline DL-RNN models LSTM, GRU, bidirectional DL-RNN models Bi-LSTM, Bi-GRU, attention-based DL-RNN models A-LSTM, A-GRU and the proposed ABD-RNN models ABi-LSTM and ABi-GRU. Table 6.1 also reports the computation time required for training each model. Here, a fixed set of h_p is considered ($E_p = 10, 50$), along with a learning rate of 0.001, $B_z = 64$, $W_z = 30$, and $O_p = \text{Adam}$.

From Table 6.1, it is clear that the proposed ABD-RNN models have outperformed all other DL-RNN models in terms of prediction accuracy for all four WQ parameters. When comparing the training time required by all DL-RNN models, it becomes evident that the baseline DL-RNN models require less training time for the same epoch. But you can note that baseline DL-RNN models take more computation time to achieve the prediction accuracy of the proposed ABD-RNN models. Table 6.1 compares the results of all DL-RNN models for $E_p = 10$ and 50 to substantiate this argument.

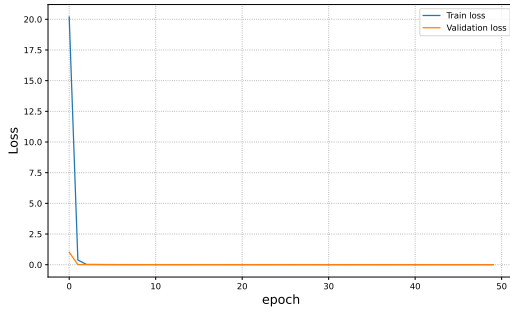
The prediction performance of salinity can be compared for $E_p = 10$ and 50. In this comparison, the prediction results of the proposed DL-RNN models with $E_p = 10$ are evaluated against all other DL-RNN models with $E_p = 50$. It is evident that for $E_p = 50$, the Bi-LSTM model with MAPE = 0.0146 demonstrates higher accuracy than the proposed DL-RNN models with $E_p = 10$. But the training time of the Bi-LSTM model with $E_p = 50$ is 8.91s, which is almost double the training time of the proposed DL-RNN models with $E_p = 10$. The proposed ABi-LSTM achieves MAPE = 0.0180 with a much lesser training time of 5.05s. And the proposed ABi-GRU achieves MAPE = 0.0170 with a training time of 5.29s, saving more than 40% computational resources required for training the model.



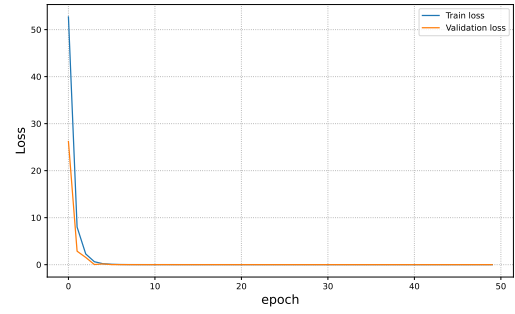
(a) LSTM



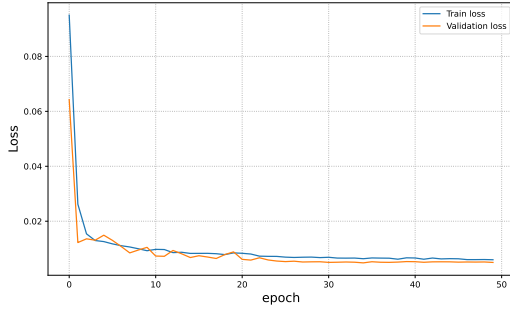
(b) GRU



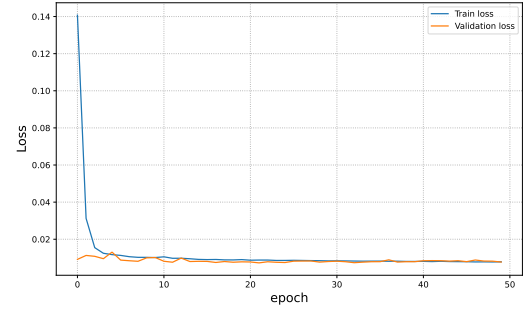
(c) Bi-LSTM



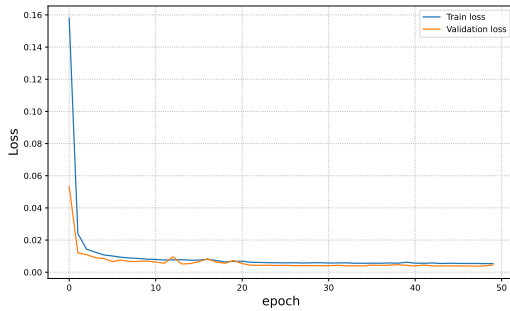
(d) Bi-GRU



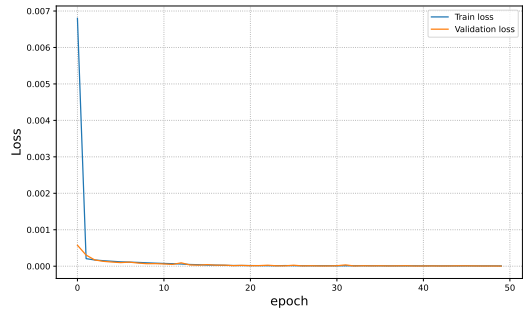
(e) A-LSTM



(f) A-GRU



(g) ABi-LSTM



(h) ABi-GRU

Figure 6.15: Loss plots for DL-RNN convergence analysis using h_p ($E_p = 50$, $L_r = 0.001$, $W_z = 30$, $B_z = 64$, and O_p with Adam) on the ADAK water quality dataset.

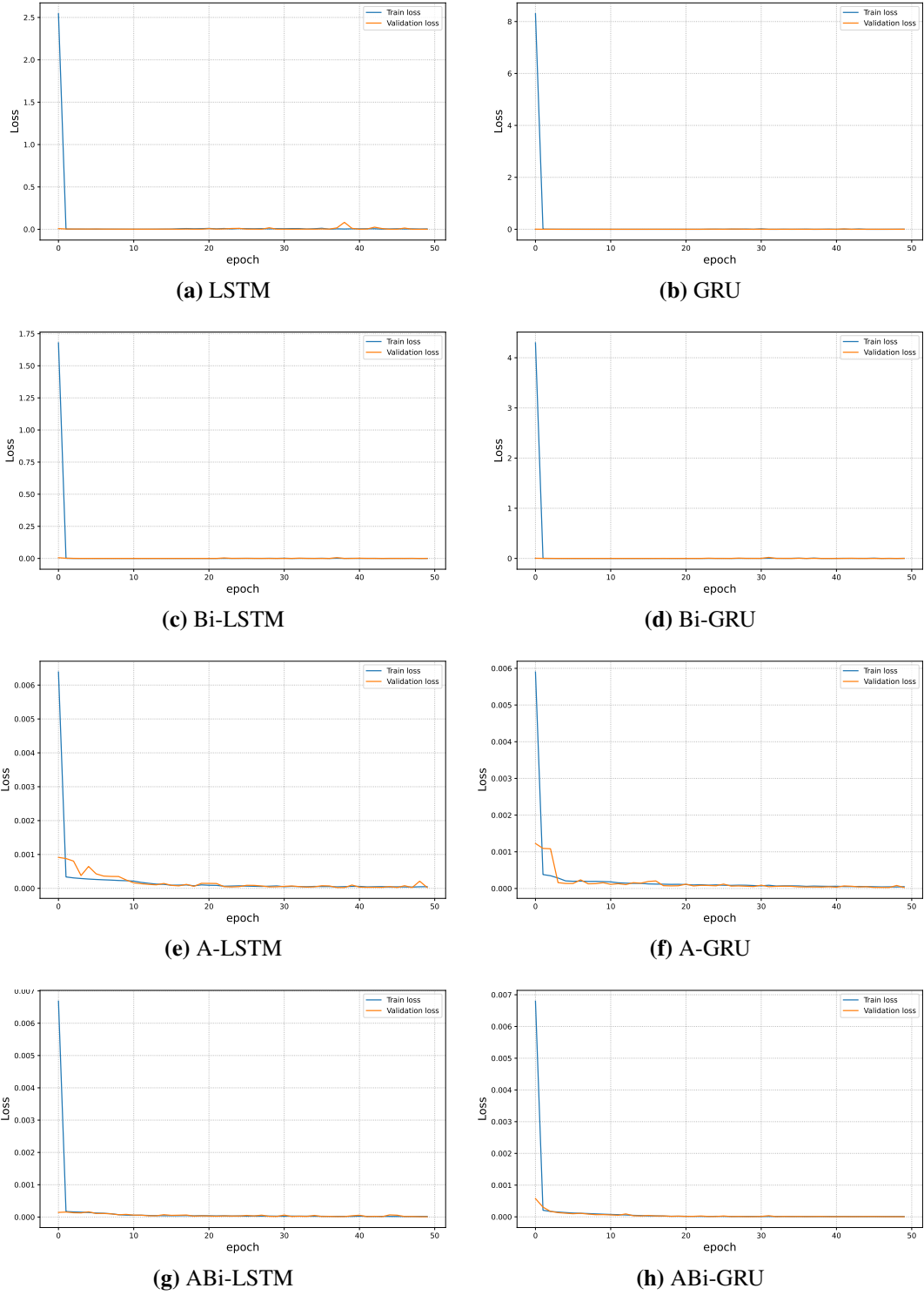


Figure 6.16: Loss plots for DL-RNN convergence analysis using h_p ($E_p = 50$, $L_r = 0.001$, $W_z = 30$, $B_z = 64$, and O_p with Adam) on the MAC water quality dataset.

Table 6.1: Prediction accuracy of LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, A-GRU, ABi-LSTM and ABi-GRU DL models for ADAK water quality dataset

Water Quality Parameter	Model	Epochs = 10					Epochs = 50				
		MAE	MSE	RMSE	MAPE	Time (s)	MAE	MSE	RMSE	MAPE	Time (s)
Salinity(ppt)	LSTM	0.3827	0.2238	0.4730	0.0247	2.82	0.3289	0.1756	0.4191	0.0212	7.01
	GRU	0.5383	0.4695	0.6852	0.0350	2.87	0.2710	0.1180	0.3435	0.0175	6.80
	Bi-LSTM	0.3157	0.1717	0.4144	0.0204	5.13	0.2257	0.0854	0.2923	0.0146	8.91
	Bi-GRU	0.3816	0.2816	0.5307	0.0245	4.35	0.2665	0.1138	0.3374	0.0171	8.75
	A-LSTM	0.2894	0.1542	0.3927	0.0186	2.83	0.2339	0.0932	0.3053	0.0151	6.71
	A-GRU	0.2705	0.1334	0.3652	0.0175	3.19	0.2580	0.1137	0.3372	0.0167	7.50
	ABi-LSTM	0.2759	0.1247	0.3532	0.0180	5.05	0.2336	0.0939	0.3064	0.0151	10.02
pH	ABi-GRU	0.2625	0.1245	0.3528	0.0170	5.29	0.2334	0.0923	0.3038	0.0152	10.31
	LSTM	0.0628	0.0064	0.0800	0.0085	2.89	0.0246	0.0011	0.0328	0.0034	6.93
	GRU	0.0355	0.0023	0.0479	0.0048	2.82	0.0330	0.0017	0.0410	0.0045	6.93
	Bi-LSTM	0.0285	0.0013	0.0355	0.0039	4.48	0.0269	0.0012	0.0348	0.0037	9.65
	Bi-GRU	0.0285	0.0014	0.0368	0.0039	4.52	0.0281	0.0013	0.0365	0.0038	9.18
	A-LSTM	0.0342	0.0017	0.0416	0.0047	2.71	0.0245	0.0010	0.0317	0.0033	7.58
	A-GRU	0.0318	0.0015	0.0389	0.0043	2.64	0.0242	0.0010	0.0318	0.0033	7.09
DO(ml/L)	ABi-LSTM	0.0272	0.0012	0.0346	0.0037	5.05	0.0255	0.0011	0.0325	0.0035	9.92
	ABi-GRU	0.0321	0.0015	0.0388	0.0044	5.09	0.0221	0.0009	0.0297	0.0030	9.88
	LSTM	0.0395	0.0025	0.0505	0.0075	2.87	0.0315	0.0017	0.0417	0.0060	6.87
	GRU	0.0379	0.0025	0.0498	0.0072	3.36	0.0248	0.0010	0.0318	0.0047	7.72
	Bi-LSTM	0.0403	0.0028	0.0534	0.0076	4.39	0.0313	0.0016	0.0399	0.0059	9.20
	Bi-GRU	0.0299	0.0014	0.0379	0.0057	4.57	0.0239	0.0010	0.0310	0.0045	9.22
	A-LSTM	0.0266	0.0011	0.0336	0.0051	2.74	0.0266	0.0011	0.0331	0.0051	6.88
Temperature(°C)	A-GRU	0.0285	0.0012	0.0353	0.0054	2.61	0.0246	0.0010	0.0321	0.0047	6.72
	ABi-LSTM	0.0268	0.0011	0.0334	0.0051	5.61	0.0240	0.0010	0.0312	0.0046	11.47
	ABi-GRU	0.0286	0.0012	0.0353	0.0054	5.17	0.0246	0.0010	0.0315	0.0047	10.23
	LSTM	0.3536	0.2130	0.4615	0.0141	3.32	0.3893	0.2640	0.5138	0.0153	7.94
	GRU	0.5352	0.4440	0.6663	0.0213	2.86	0.3217	0.1881	0.4337	0.0127	7.42
	Bi-LSTM	0.2752	0.1187	0.3445	0.0109	4.23	0.2699	0.1161	0.3408	0.0107	10.68
	Bi-GRU	0.2814	0.1270	0.3564	0.0112	4.72	0.2431	0.0976	0.3124	0.0096	10.46
	A-LSTM	0.2509	0.0990	0.3147	0.0100	2.83	0.2437	0.0939	0.3065	0.0097	8.72
	A-GRU	0.2550	0.1088	0.3299	0.0101	2.70	0.2427	0.0962	0.3101	0.0096	7.79
	ABi-LSTM	0.2376	0.0889	0.2982	0.0094	5.59	0.2432	0.0958	0.3095	0.0096	10.94
	ABi-GRU	0.2410	0.0938	0.3062	0.0095	5.12	0.2467	0.0992	0.3150	0.0098	10.55

$h_p : E_p = \{10, 50\}$, $L_r = 0.001$, $W_z = 30$, $B_z = 64$, and Adam optimizer.

Further, the prediction performance of pH for $E_p = 10$ and 50 is compared. It can be observed that for $E_p = 50$, the A-LSTM DL-RNN model with MAPE = 0.0033 has better accuracy than the proposed DL-RNN models with $E_p = 10$. But with a training time of 7.58s. The proposed ABi-LSTM achieves MAPE = 0.0037 with a much lesser training time of 5.05s. And the proposed ABi-GRU achieves MAPE = 0.0044 with a training time of 5.09s, saving almost 25% of the computational resource required for training the model.

When making a similar comparison for DO, it is observed that the Bi-GRU model with $E_p = 50$ exhibits better prediction performance compared to the proposed models with $E_p = 10$. The Bi-GRU model achieves a MAPE of 0.0045, whereas the proposed model achieves a MAPE of 0.0051 for $E_p = 10$, with a computation time of only 5.61s. However, the computation time for the Bi-GRU DL-RNN model is 9.22s, resulting in a 40% reduction in computational resources required for training the model. Furthermore, when comparing the prediction performance for temperature, it is observed that the A-GRU model with $E_p = 50$ closely matches the performance of the proposed DL-RNN models with $E_p = 10$. The A-GRU models achieve a MAPE of 0.0096 with

a training time of 7.79s. In comparison, the proposed models achieve a slightly better MAPE of 0.0094 with a training time of 5.59s, thereby saving almost 30% of computational resources. Based on the information provided in Table 6.1, it can be concluded that the proposed ABD-RNN models outperform other DL-RNN models in terms of prediction accuracy while requiring fewer computational resources.

6.4.2 MAC water quality dataset

The prediction performance of the MAC WQ dataset is reported in Table 6.2, which gives MAE, MSE, RMSE and MAPE comparative analysis of the proposed ABD-RNN models (ABi-LSTM and ABi-GRU) with baseline DL-RNN models (LSTM and GRU), bidirectional DL-RNN models (Bi-LSTM and Bi-GRU), and attention-based DL-RNN models (A-LSTM and A-GRU). Table 6.2 also reports the computation time required for training each model in seconds. Here, a fixed set of h_p is considered ($E_p = 10, 50$), along with a learning rate of 0.001, $B_z = 64$, $W_z = 30$, and $O_p = \text{Adam}$.

Table 6.2: Prediction accuracy of LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, A-GRU, ABi-LSTM and ABi-GRU DL models for MAC water quality dataset

Water Quality Parameter	Model	Epochs = 10					Epochs = 50				
		MAE	MSE	RMSE	MAPE	Time (s)	MAE	MSE	RMSE	MAPE	Time (s)
Salinity(ppt)	LSTM	0.1578	0.1737	0.4168	0.0047	24.62	0.1167	0.1081	0.3288	0.0035	108.87
	GRU	0.1011	0.1057	0.3251	0.0030	33.30	0.3251	0.6111	0.7817	0.0096	103.87
	Bi-LSTM	0.0984	0.0402	0.2005	0.0029	62.73	0.0358	0.0077	0.0876	0.0011	143.78
	Bi-GRU	0.2677	1.7427	1.3201	0.0083	52.17	0.2514	0.4023	0.6343	0.0075	163.78
	A-LSTM	0.1898	0.1822	0.4268	0.0057	37.63	0.1010	0.0314	0.1771	0.0030	147.60
	A-GRU	0.1223	0.0651	0.2551	0.0036	30.71	0.0773	0.0255	0.1598	0.0023	117.75
	ABi-LSTM	0.0625	0.0153	0.1237	0.0018	60.70	0.0263	0.0078	0.0882	0.0008	163.99
	ABi-GRU	0.0709	0.0151	0.1228	0.0021	50.08	0.0223	0.0038	0.0617	0.0007	149.01
pH	LSTM	0.0258	0.0033	0.0574	0.0030	30.80	0.0067	0.0001	0.0102	0.0008	109.21
	GRU	0.1119	0.1150	0.3392	0.0127	28.23	0.1871	0.1310	0.3619	0.0217	103.39
	Bi-LSTM	0.0174	0.0007	0.0256	0.0020	53.47	0.0122	0.0004	0.0205	0.0014	254.55
	Bi-GRU	0.0221	0.0034	0.0581	0.0025	45.57	0.0341	0.0123	0.1111	0.0039	215.89
	A-LSTM	0.0151	0.0006	0.0244	0.0018	32.96	0.0111	0.0005	0.0216	0.0013	166.99
	A-GRU	0.0153	0.0007	0.0269	0.0018	27.60	0.0104	0.0004	0.0210	0.0012	148.13
	ABi-LSTM	0.0120	0.0005	0.0226	0.0014	55.51	0.0089	0.0004	0.0198	0.0010	243.47
	ABi-GRU	0.0117	0.0007	0.0262	0.0014	35.14	0.0087	0.0005	0.0232	0.0010	158.24
DO(ml/L)	LSTM	0.8591	1.7852	1.3361	0.2111	23.52	0.1778	0.0544	0.2332	0.0496	110.41
	GRU	0.2744	0.4710	0.6863	0.0503	22.57	0.1829	0.1879	0.4335	0.0384	101.92
	Bi-LSTM	0.3547	0.7464	0.8639	0.0601	43.22	0.1138	0.0308	0.1754	0.0271	248.23
	Bi-GRU	0.6842	4.1303	2.0323	0.0920	34.57	0.4389	1.9743	1.4051	0.0524	176.40
	A-LSTM	0.6402	1.0682	1.0336	0.1486	30.34	0.1875	0.1154	0.3397	0.0419	158.22
	A-GRU	0.2174	0.2372	0.4870	0.0369	30.88	0.1642	0.1405	0.3749	0.0296	114.72
	ABi-LSTM	0.2088	0.1667	0.4083	0.0435	69.19	0.1577	0.0739	0.2719	0.0404	150.57
	ABi-GRU	0.1335	0.1081	0.3288	0.0237	59.26	0.1062	0.1090	0.3301	0.0157	178.27
Temperature($^{\circ}\text{C}$)	LSTM	0.7791	0.7409	0.8608	0.0284	32.62	1.0359	1.0853	1.0418	0.0379	116.69
	GRU	0.1330	0.0390	0.1974	0.0048	27.19	0.4753	0.2287	0.4782	0.0174	119.78
	Bi-LSTM	0.1445	0.0489	0.2212	0.0053	56.01	0.0885	0.0087	0.0934	0.0032	246.85
	Bi-GRU	0.0776	0.0122	0.1103	0.0028	45.06	0.0614	0.0042	0.0651	0.0022	193.02
	A-LSTM	0.0800	0.0143	0.1194	0.0029	35.81	0.0532	0.0070	0.0839	0.0019	130.40
	A-GRU	0.1082	0.0294	0.1716	0.0039	30.23	0.1042	0.0176	0.1325	0.0038	108.78
	ABi-LSTM	0.0774	0.0152	0.1232	0.0028	56.36	0.0297	0.0032	0.0565	0.0011	200.36
	ABi-GRU	0.0803	0.0163	0.1278	0.0029	50.13	0.0204	0.0025	0.0497	0.0007	167.30

$h_p : E_p = \{10, 50\}$, $L_r = 0.001$, $W_z = 30$, $B_z = 64$, and Adam optimizer.

From Table 6.2, it is clear that the proposed ABD-RNN models have outperformed all other DL-RNN models in terms of prediction accuracy for all four WQ parameters. When comparing the training time required by all DL-RNN models, it becomes evident

that the baseline DL-RNN models need less training time for the same epoch. It can be noted that baseline DL-RNN models take more computation time to achieve the prediction accuracy of the proposed ABD-RNN models. Table 6.2 compares the results of all DL-RNN models for E_p 10 and 50 to substantiate this argument.

The prediction performance of salinity for $E_p = 10$ and 50 is compared. In this comparison, the prediction results of proposed DL-RNN models with $E_p = 10$ and the prediction results of all other DL-RNN models with $E_p = 50$. It is evident that for $E_p = 50$, the Bi-LSTM model with MAPE = 0.0011 has better accuracy than the proposed DL-RNN models with $E_p = 10$. But the training time of the Bi-LSTM model with $E_p = 50$ is 143.78s, which is more than double the training time of the proposed DL-RNN models with $E_p = 10$. The proposed ABi- LSTM achieves MAPE = 0.0018 with a much lesser training time of 60.70s. And the proposed ABi-GRU achieves MAPE = 0.0021 with an even lesser training time of 50.08s, saving more than 50% computational resources required for training the model.

Further, the prediction performance of pH for $E_p = 10$ and 50 is compared. it can be observed that for $E_p = 50$, the LSTM DL-RNN model with MAPE = 0.0008 has better accuracy than the proposed DL-RNN models with $E_p = 10$ and 50. But with a training time of 109.21s, almost double the training time of the proposed DL-RNN models for $E_p = 10$. Another DL-RNN model is A-GRU achieves a MAPE = 0.0012 but requires a training time of 148.13s. The proposed ABi- LSTM achieves MAPE = 0.0014 with a much lesser training time of 55.51s. And the proposed ABi-GRU achieves MAPE = 0.0014 with an even lesser training time of 35.14s, saving more than 65% of the computational resource required for training the model.

When making a similar comparison for DO, it can be observed that Bi-LSTM with $E_p = 50$ is the only model that comes close to the prediction performance of proposed models for $E_p = 10$. Here Bi-LSTM has a MAPE = 0.0271, whereas the proposed model achieves a MAPE = 0.0237 for $E_p = 10$ and with just a computation time of 59.26s. But the computation time of the Bi-LSTM DL-RNN model is 248.23s which is almost four times the computation time required by proposed models, saving 75% of computational resources for training the model. When comparing the prediction performance for temperature, it is observed that the Bi-GRU model with $E_p = 50$ comes close to the prediction performance of the proposed DL-RNN models with $E_p = 10$. The Bi-GRU models achieve MAPE = 0.0022 but with a training time of 193.02s. In comparison, the proposed models achieve MAPE = 0.0029 with a training time of just 56.36s saving almost 70% computational resources. From Table 6.2, it can be concluded that the proposed ABD-RNN models have outperformed all other DL-RNN models in prediction accuracy and use fewer computational resources.

In the evaluation of WQP models on the ADAK and MAC datasets, the proposed ABD-RNN models, namely ABi-LSTM and ABi-GRU, consistently outperformed all other baseline and attention-based DL-RNN models in terms of prediction accuracy for various water quality parameters, including salinity, pH, DO, and temperature. The ABD-RNN models achieved lower MAE, MSE, RMSE, and MAPE values, indicating their superior predictive capabilities. Despite requiring slightly longer training times, the ABD-RNN models demonstrated significant advantages in computational efficiency compared to some baseline models, making them an attractive choice for accurate water quality prediction tasks. This performance improvement signifies the potential of ABD-RNN models in enhancing water quality monitoring and management systems.

6.5 SUMMARY

This chapter introduces the ABD-RNN approach (ABD-RNN) with Adam optimization, aimed at enhancing the accuracy of aquaculture WQP. The proposed models incorporate bidirectional DL-RNN layers to capture the sequence correlation of the data effectively, while the attention mechanism assigns varying weights to focus on crucial data points. A comprehensive investigation is conducted on various DL-RNN models to assess the impact of different hyperparameters. Based on this study, optimal values for h_p are determined and utilized for performance comparison and further analysis in the results section. The outcomes demonstrate that the proposed ABD-RNN models, namely ABi-LSTM and ABi-GRU, surpass other DL-RNN models, including LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, and A-GRU, in terms of both prediction accuracy and computational efficiency. These findings emphasize the effectiveness of the ABD-RNN approach for aquaculture WQP tasks.